MATH 420 HW4 Sadullaev L.

Problem 1.

For each part justify you claim as to whether or not S is a basis for the given set?

Solution:

(a)

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ -4 & 4 \end{bmatrix} \right\} : \text{Is } S \text{ a basis for } M_{2\times 2}(\mathbb{R})?$$

(i) Recalling the theorem about dimension & basis, since $M_{2\times 2}(\mathbb{R})$ has 4 basis vectors, S matches this criteria. The only thing I'm going to see is if S is L.I.?

(ii)

$$a\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} + b\begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} + c\begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 2 & 1 \\ -4 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad (1)$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & -1 & 2 & 1 \\ -2 & 1 & 1 & -4 \\ 1 & 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad (2)$$

$$\text{rref} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & -1 & 2 & 1 & 0 \\ -2 & 1 & 1 & -4 & 0 \\ 1 & 1 & 0 & 4 & 0 \end{array}\right] = \left[\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & -2 & -1 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$\text{rank}(A) = 3 < 4 \Rightarrow (1)$ has infinitely many solutions

So S is L.D. $\Rightarrow$ S fails to be a basis for $V = M_{2\times 2}(\mathbb{R})$.

(b) Is $S = \left\{ \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix} \right\}$ a basis for $M_{2\times 2}(\mathbb{R})$?

(i) $|S| = 4$ (cardinality of S) matches $\dim M_{2\times 2}(\mathbb{R})$. (1)

(ii) L.I. for S:

$$a\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} + b\begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} + c\begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (2)$$

$$\text{rref}\left[\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & -1 & 2 & 1 & 0 \\ -2 & 1 & 1 & 2 & 0 \\ 1 & 1 & 0 & -2 & 0 \end{array}\right] = \left[\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & -2 & -1 & 0 \\ 0 & 0 & 1 & 7 & 0 \\ 0 & 0 & 0 & 24 & 0 \end{array}\right] \quad (3)$$

rank $(A) = 4 \Rightarrow$ (2) has only **trivial solution.** $\Rightarrow$

S is L.I. $\Rightarrow$ considering (i) & (ii) $\Rightarrow S$ is a basis.

(c)

Is $S = \{(x^3+2x^2), (-x^2+3x+1), (x^3-x^2+2x-1), (1+x)\}$ a basis for P_3?

(i) $|S| = 4 = \dim(P_3)$ holds (1)

(ii) L.I. for S:

$$a(x^3+2x^2) + b(-x^2+3x+1) + c(x^3-x^2+2x-1) + d(1+c) = 0 \quad (2)$$

$$\begin{cases} a+c=0 \\ 2a-b-c=0 \\ 3b+2c+d=0 \\ b-c+d=0 \end{cases} \xrightarrow[\text{solution}]{\text{has unique}} \begin{cases} a=0 \\ b=0 \\ c=0 \\ d=0 \end{cases} \Rightarrow S \text{ is L.I.} \quad (3)$$

Considering (1) & (3), S is a basis for P_3.

(d) Is $S = \{(x^3 - x), (2x^2 + 4), (-2x^3 + 3x^2 + 2x + 6), (x+2), (x^2 + 2x + 3)\}$ a basis for P_3?

(i) $|S| = 5 \neq 4 = \dim(P_3)$ (1)

(ii) S has 5 vectors in 4 dimensional space P_3 so, S can't be L.I. (2)

S fails in (i) & (ii) to be a basis for P_3.

Problem 2.

Let u, v, w be distinct vectors of a vector space V. Show that if $\{u, v, w\}$ is a basis for V, then $\{u, u+v, u+v+w\}$ is also a basis for V.

Solution:

Let $u \neq v \neq w$ be vectors of a vector space V.

$\Rightarrow u \neq v \neq w \in V$ (1)

Assume $\beta = \{u, v, w\}$ is a basis for V. (2)

WTS: $\gamma = \{u, u+v, u+v+w\}$ is also a basis for V:

Since β is a basis for V, $V = \text{span}(\beta)$. $\Rightarrow$

$\Rightarrow \forall x \in V, \ \exists a_1, a_2, a_3 \in F \ni$

$$x = a_1 u + a_2 v + a_3 w \qquad (3)$$

But now let $\begin{cases} a_3 = c_3 \\ a_2 = c_2 + c_3 \\ a_1 = c_1 + c_2 + c_3 \end{cases}$ (4) for another group of $c_1, c_2, c_3 \in F$ scalars.

Substituting (4) into (3), we get:

$$(c_1 + c_2 + c_3) u + (c_2 + c_3) v + c_3 w = x, \quad \forall x \in V$$

$$\underline{c_1 u} + \underline{\underline{c_2 u}} + \underset{\sim}{c_3 u} + \underline{\underline{c_2 v}} + \underset{\sim}{c_3 v} + \underset{\sim}{c_3 w} = x, \quad \forall x \in V$$

$$c_1 u + c_2 (u + v) + c_3 (u + v + w) = x \qquad (5)$$

$\forall x \in V$.

(5) is sure in $\text{span}(\gamma)$.

Because x, an arbitrary point from $V = \text{span}(\beta)$, belongs to $\text{span}(\gamma)$, it follows that

$$\text{span}(\beta) \subseteq \text{span}(\gamma) \qquad (6)$$

Now let me show the reverse inclusion:

$\text{span}(\beta) \supseteq \text{span}(\gamma)$

(2) Let $y \in \text{span}\,\gamma = \text{span}(\{u, u+v, u+v+w\})$.

Then $\exists$ scalars $c_1, c_2, c_3 \in F \ni$

$$c_1 u + c_2(u+v) + c_3(u+v+w) = y \quad (7)$$

$$\Rightarrow (c_1+c_2+c_3)u + (c_2+c_3)v + c_3 w = y \quad (8)$$

Obviously, (8) is from $\text{span}(\beta) \Rightarrow y \in \text{span}(\beta)$

$$\Rightarrow \quad \text{span}(\gamma) \subseteq \text{span}(\beta) \quad (9)$$

(6) & (9) help us conclude $\text{span}(\gamma) = \text{span}(\beta) = V$. (10)

But now the only thing is to show γ is L.I when β is a basis.

Since β is a basis, it is L.I. $\Rightarrow$

$$a_1 u + a_2 v + a_3 w = 0 \quad (11)$$

only when $a_1 = a_2 = a_3 = 0$ (12)

Given that, let's check $\mathcal{J}$ for L.I. relation

$$b_1 \underline{u} + b_2 (\underline{u} + \underline{v}) + b_3 (\underline{u} + \underline{v} + w) = 0 \quad (13)$$

$$\Rightarrow (b_1 + b_2 + b_3) u + (b_2 + b_3) v + b_3 w = 0 \quad \Rightarrow$$

$$\begin{cases} b_1 + b_3 + b_3 = 0 \\ b_2 + b_3 = 0 \\ b_3 = 0 \end{cases} \xrightarrow[\text{solution}]{\text{unique}} b_1 = b_2 = b_3 = 0 \quad (14)$$

(14) shows that $\mathcal{J}$ is L.I. $\Rightarrow$

So, $\mathcal{J}$ is a spanning L.I. set for $V \Rightarrow$

$\mathcal{J}$ is a basis.

Problem 3

Prove that a vector space is infinite dimensional if and only if it contains an infinite L.I. subset.

Solution:

Let me denote the following notations:

P = "A vector space is infinite dimensional"

Q = "It contains a L.I subset with infinite vectors".

WTS: $P \Leftrightarrow Q$. (1)

Yet, instead of (1), we show Not $P \Leftrightarrow$ Not Q (2)

So,

Not P = "A vector space is NOT infinite dimensional" $\Rightarrow$

Not P = "A vector space is FINITE dimensional". (3)

Not Q = "It DOES NOT contain an infinite L.I subset" $\Rightarrow$

Not Q = "It contains ONLY FINITE L.I. subsets". (4)

($\Leftarrow$) in (2) Let Not P holds. $\Rightarrow$
A vector space is finite dimensional. $\Rightarrow$

Theorem 1.9.

If a vector space V is generated by a finite set S, then some subset of S is a basis for V. Hence V has a finite basis.

Using this Th 1.9. V has a finite basis $\beta = \{u_1, u_2, \dots, u_n\}$. $\Rightarrow$

$\dim(V) = |\beta| = n$. (5)

Hence, $\forall$ L.I subset W, $\dim W \leq n$, because if $\dim W > n$, then W is L.D, thus, $\forall$ L.I $W \subseteq V$ is finite dimensional.

$\boxtimes$

($\Leftarrow$)

Let not Q hold in (2) $\Rightarrow$

Let me bring corollary from book:

Corollary 2 (For replacement Th. 1.10.)

Let V be a vector space with dimen. n.

(a) Any finite generating set for V contains at least n vectors, and a generating set for V that contains exactly n vectors is a basis for V.

(b) Any L.I. subset of V that contains exactly n vectors is a basis for V

(c) Every L. I. subset of V can be extended to a basis for V.

Not Q holds means $\forall$ L.I. $W \subseteq V$ is finite. Then by the Corollary 2 part (c) (which I highlighted more), W can be extended to a basis β with n elements, hence $\dim(V) = |\beta| = n \Rightarrow$

V is finite dimensional.

$\blacksquare$

Problem 4.

Let V be a finite dimensional vector space over $\mathbb{C}$ with dimension n. Prove that if V is now regarded as a vector space over $\mathbb{R}$, then $\dim(V) = 2n$.

Solution:

Let V be an n dimensional vector space over $\mathbb{C}$. Then

$\forall x \in V$, $\exists$ scalars $c_1, c_2, \ldots c_n \in \mathbb{C}$ $\ni$

$$c_1 u_1 + c_2 u_2 + \ldots + c_n u_n = x \qquad (1)$$

where $\beta = \{u_1, u_2, \ldots u_n\}$ is a basis for V.

Now if we consider V over $\mathbb{R}$, for the same vector above, we need $c_j = a_j + i b_j$ substitution and

$$(a_1 + i b_1) u_1 + (a_2 + i b_2) u_2 + \dots + (a_n + i b_n) u_n = x \qquad (2)$$

$\forall x \in V$

since $i \notin \mathbb{R}$, we need to include i in our vectors u_j to make the coefficients real:

$$a_1 u_1 + b_1 \overbrace{(i u_1)}^{\tilde{u}_1} + a_2 u_2 + b_2 \overbrace{(i u_2)}^{\tilde{u}_2} + \dots + a_n u_n + b_n \overbrace{(i u_n)}^{\tilde{u}_n} = x$$

$$\Rightarrow a_1 u_1 + b_1 \tilde{u}_1 + a_2 u_2 + b_2 \tilde{u}_2 + \dots + a_n u_n + b_n \tilde{u}_n = x \qquad (3)$$

So, now $\forall x \in V$, to write it as a linear combination of the vectors in V, we need $\tilde{\beta} = \{u_1, \tilde{u}_1, u_2, \tilde{u}_2 \dots u_n, \tilde{u}_n\}$ new basis. $\Rightarrow$

$$|\tilde{\beta}| = \dim V = 2n \qquad (4)$$

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